

Chunyang Wang

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WORK EXPERIENCE

Project Researcher

11/2025 - Now

National Institute of Informatics

- Host: Yuichi Yoshida

EDUCATION

Ph.D in Computer Science

09/2020 - 07/2025

Nanjing University

- Field: Theoretical Computer Science
- Advisor: Yitong Yin

B.Sc in Computer Science

09/2016 - 07/2020

Nanjing University

RESEARCH INTERESTS

- Counting and sampling algorithms
- Discrete probability
- Algorithmic stability

SELECTED AWARDS & HONORS

The 45th International Collegiate Programming Contest, Shenyang Regional

2021-07-18

2nd Place, Gold Medal (Teammates: Tianxing Ding, Qicheng Shan)

Liaoning, China

The 47th International Collegiate Programming Contest, Nanjing Regional

2022-12-18

2nd Place, Gold Medal (Teammates: Hongyang Liu, Xinyuan Zhang)

Jiangsu, China

The 47th International Collegiate Programming Contest, World Finals

2024-04-15

6th Place, Silver Medal (Teammates: Hongyang Liu, Xinyuan Zhang)

Luxor, Egypt

National Scholarship (2023)

2023-10-12

PUBLICATIONS

1. Kun He, Chunyang Wang, Yitong Yin.
Sampling Lovász local lemma for general constraint satisfaction solutions in near-linear time.
In **FOCS** 2022: 147-158.
2. Kun He, Chunyang Wang, Yitong Yin.
Deterministic counting Lovász local lemma beyond linear programming.
In **SODA** 2023: 3388-3425.
3. Weiming Feng, Heng Guo, Chunyang Wang, Jiaheng Wang, Yitong Yin.
Towards derandomising Markov chain Monte Carlo.
in **SICOMP** 2025: 775-813 (preliminary version in **FOCS** 2023: 1963-1990)
4. Chunyang Wang, Yitong Yin.
A Sampling Lovász Local Lemma for Large Domain Sizes.
in **FOCS** 2024: 129-150
5. Aditya Lonkar, Zongchen Chen, Chunyang Wang, Kuan Yang, Yitong Yin.
Counting random k -SAT near the satisfiability threshold.
in **STOC** 2025: 867-878
6. Jingcheng Liu, Chunyang Wang, Yixiao Yu, Yitong Yin.
Phase Transitions via Complex Extensions of Markov Chains.
in **STOC** 2025: 903-914

7. Konrad Anand, Graham Freifeld, Heng Guo, Chunyang Wang, Jiaheng Wang.
Sink-free orientations: a local sampler with applications
in **RANDOM** 2025: 60:1–60:19
8. Hongyang Liu, Chunyang Wang, Yitong Yin.
Local Gibbs sampling beyond local uniformity.
in **SODA** 2026: 996–1025

PREPRINTS

1. Chunyang Wang, Yuichi Yoshida, Zihan Zhang.
Approximate Colorwise Tensorization of Entropy and Optimal Mixing of the Wang-Swendsen-Kotecký Dynamics.
arXiv:2607.10119
2. Hongyang Liu, Chunyang Wang, Yitong Yin, Yiyao Zhang, Can Zhou.
A Counting Lovász Local Lemma.
arXiv:2608.08616

TALKS

Sampling and Counting Solutions to Constraint Satisfaction Problems.
NCTCS, Zhejiang Normal University, Jinhua, Zhejiang, China, Aug. 2026

Local Gibbs Sampling beyond Local Uniformity.
NII Algorithm Lunch Seminar, National Institute of Informatics, Tokyo, Japan, Nov. 2025

SODA, Vancouver, British Columbia, Canada, Jan. 2026

Counting random k -SAT near the satisfiability threshold.
LFCS Seminar, University of Edinburgh, Edinburgh, UK, Dec. 2024

Complexity & Algorithms Workshop, Sun Yat-sen University, Guangzhou, Guangdong, China, Apr. 2025

STOC, Prague, Czech Republic, Jun. 2025

A Sampling Lovász Local Lemma for Large Domain Sizes.
ACO Student Seminar, Georgia Institute of Technology, Atlanta, Georgia, USA, Oct. 2024

FOCS, Chicago, Illinois, USA, Oct. 2024

Student Theory Day, University of Science and Technology of China, Hefei, Anhui, China, Mar. 2025

Towards derandomising Markov chain Monte Carlo.
MCMC 2.0 Workshop, NII Shonan Meeting, Hayama, Kanagawa, Japan, Sep. 2023

FOCS, Santa Cruz, California, USA, Nov. 2023

A sampling Lovász local lemma.
2023 CCF Forum for Distinguished Ph.D. Candidates in Theoretical Computer Science, Hong Kong, China, Jul. 2023

Deterministic counting Lovász local lemma beyond linear programming.
SODA (virtual presentation), Florence, Italy, Jan. 2023

TEACHING ASSISTANTSHIPS

<i>Advanced Algorithms (Fall 2020)</i> Instructor: Yitong Yin	Nanjing University
<i>Advanced Algorithms (Fall 2022)</i> Instructor: Yitong Yin	Nanjing University
<i>Combinatorics (Spring 2023)</i> Instructor: Yitong Yin	Nanjing University
<i>Combinatorics (Spring 2024)</i> Instructor: Yitong Yin	Nanjing University

SERVICES

Reviewing:

Conference: COCOON 2024, TAMC 2024, STOC (2024, 2025, 2026), ITCS 2025, FOCS 2026, ICALP 2026, ISAAC 2026, SODA 2027

Journal: TALG 2024